1. Transport in Mammals

1.1. The Circulatory System

- The Mammalian Circulatory System is a **closed double-circulation**
- 'Closed' because blood is contained within blood vessels
- 'Double' because blood passes through the heart twice in one circulation of the body (pulmonary circuit and systemic circuit)
- The circulatory system includes a heart and blood vessels (arteries, arterioles, capillaries, venules and veins)

Structure of arteries and veins:

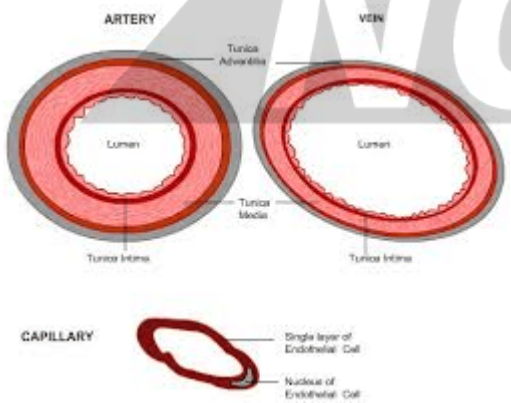
- Three layers of tissues known as the tunica intima, tunica media, and tunica externa
- **Tunica interna:** Endothelium, made of squamous epithelial cells
- **Tunica media:** In arteries- elastic fibres, collagen fibres and smooth muscle. In veins- smooth muscle and elastic fibres
- **Tunica externa:** In arteries, collagen fibres and some elastic fibres. In veins- mostly collagen fibres
- Arteries have a narrow lumen; veins have a wide lumen

Role of elastic fibres, smooth muscle, and collagen:

- **Elastic fibres:** Recoil and contract, squeezing the blood and so moving it along in a continuous flow; they allow the walls to stretch as pulses of blood surge through
- **Smooth muscles:** Contract to reduce and relax to increase blood flow; this controls the volume of blood flowing into a tissue
- **Collagen:** Gives strength, structure and flexibility

Structure of Capillaries:

- Wall made of endothelium (one cell thick)
- Lumen is about 7μm wide



Some of the main blood vessels and their functions:

- **Pulmonary artery:** transports deoxygenated blood from the right ventricle to the lungs
- **Pulmonary vein:** transports oxygenated blood from the lungs to the left atrium
- **Aorta (artery):** transports oxygenated blood from the left ventricle to the body
- **Vena cava (vein):** transports deoxygenated blood from the body to the right atrium

1.2. Blood Vessels

Arteries:

- These blood vessels transport oxygenated blood swiftly to the tissues at high pressures
- They have thick walls to withstand this high-pressure
- The exception to this is the pulmonary artery, carrying deoxygenated blood to the lungs
- Elastic arteries are closer to the heart than the final destination and have greater proportion of elastic fibres to withstand the higher pressure of blood
- Muscular arteries are closer to the final destination than the heart and have greater proportion of smooth muscle to adjust the diameter and, hence the blood flow
- As arteries reach tissues, they branch into smaller vessels called arterioles

Capillaries:

- Arterioles continue to branch into capillaries
- These are the smallest blood vessels, and they take the blood as close as possible to the cells
- This allows for rapid transfer of substances between cells and the blood
- Due to the very small diameter of these blood vessels, blood travels very slowly
- This increases the opportunity for diffusion to occur
  - Walls of capillaries are made of a single layer of endothelial cells with pores between individual cells present to allow some components of blood to pass through into the cells and tissues of the body

Veins:

- Many capillaries join to form venules, which then join to form veins
- These blood vessels carry deoxygenated blood back to the heart
- The exception is a pulmonary vein, carrying oxygenated blood from the lungs to the heart
  - Tunica media is thinner in veins when compared to arteries, as the blood in veins is at a lower pressure
  - Valves are also present to prevent backflow of blood
  - To keep blood flowing upwards from the legs, veins are usually near muscles, so pressure is increased when muscles contract

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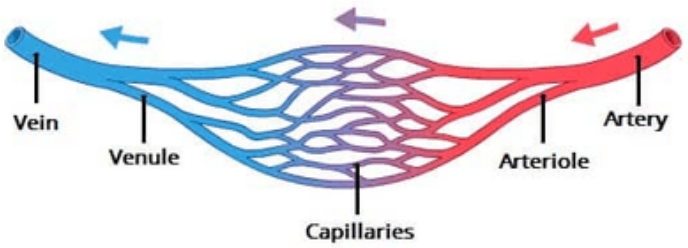


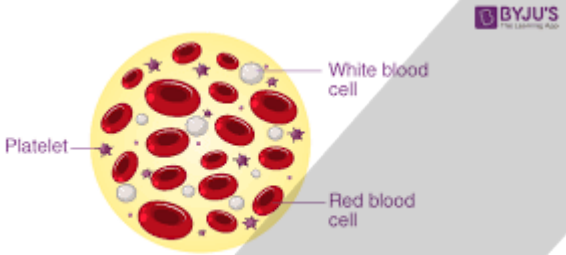
1.3. Blood and Tissue Fluid

Water is the main component of blood and tissue fluid. It has a high specific heat capacity and provides stable temperature conditions for the fluids, making them ideal for the cells and tissues inside.

- It is the universal solvent and can dissolve most substances, making it an ideal blood and tissue fluid component.

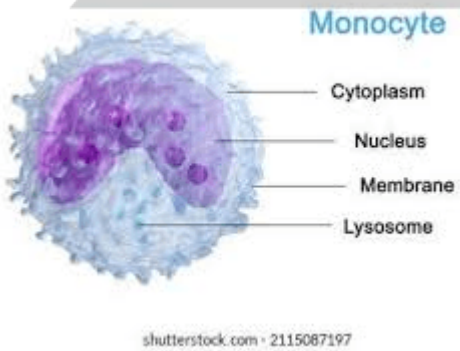
Blood is composed of **4 components**

- **Plasma**:
  - This is the liquid part of the blood
  - It is a dilute solution of salts, glucose, amino acids, vitamins, urea, protein and fats
- **Platelets**: Involved in blood clotting
- **Red blood cells**: Involved in carrying oxygen
- **White blood cells:** Involved in the immune system

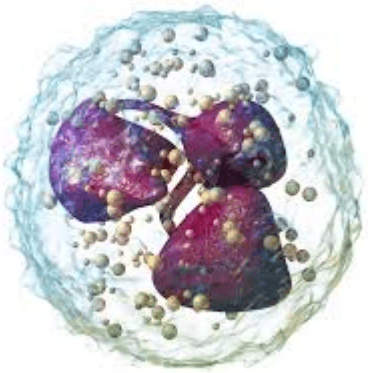


Three kinds of **white blood cells**:

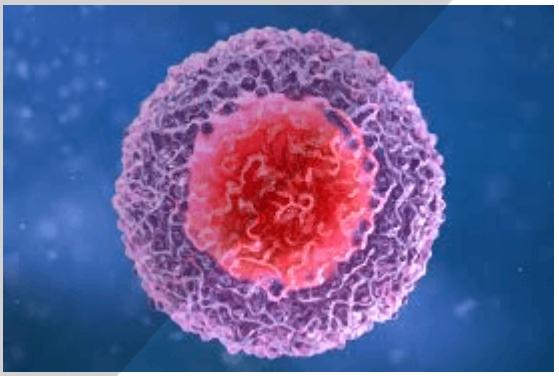
- **Monocytes:** Largest type of WBCs, act as phagocytes, and can differentiate into macrophages, kidney-shaped nucleus



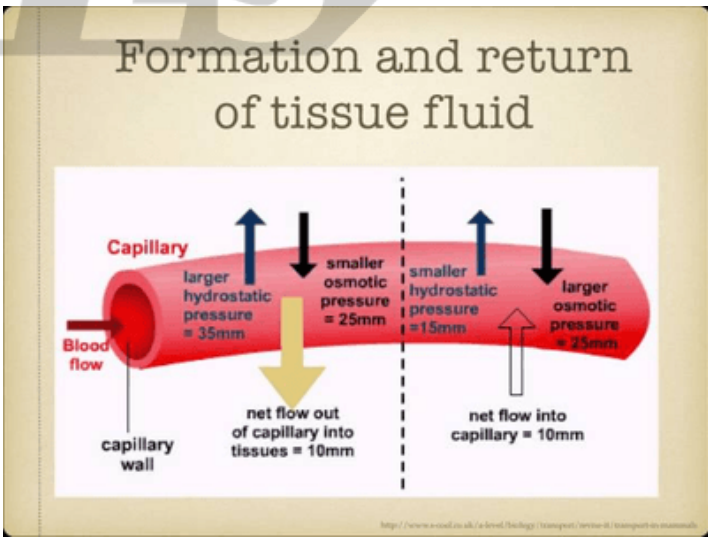
- **Neutrophils**: Can act as phagocytes, lobed nuclei and granular cytoplasm



- **Lymphocyte**: Produces antibodies against specific antigens, the circular nucleus that covers almost all the cytoplasm.



- **Tissue Fluid**
  - This fluid surrounds all the cells.
  - Substances move from the blood to the tissue fluid, and from the tissue fluid, they diffuse into the cells.
  - Facilitates substance exchange between cells and the blood by helping bring oxygen and nutrients to cells and removing waste products from them as well
  - Tissue fluid has almost the same components as plasma but lacks large plasma proteins, which are too large to diffuse through pores in capillaries.
  - Osmotic pressure causes tissue fluid to move into and out of capillaries.



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1.4. Transport of Oxygen and Carbon Dioxide

Haemoglobin (Hb) (in the RBCs) transports oxygen and carbon dioxide:

In Respiring Tissues:

- The pCO_2 is high and the pO_2 is low
- CO2 from the cells diffuses into the plasma
- CO2 combines with the -NH2 terminal of Hb to form carbaminohaemoglobin (10% carried this way)
- 5% of CO2 directly dissolves in the plasma and is transported
- Most CO2 combines with water (catalysed by carbonic anhydrase) to form carbonic acid, which then dissociates into H+ and HCO3- ions (85%)
- H+ ions combine with Hb to form Haemoglobinic acid (HHb) (This prevents blood from becoming too acidic)

$$\underset{\text{carbon dioxide}}{CO_2} + \underset{\text{water}}{H_2O} \xrightleftharpoons{\text{carbonic anhydrase}} \underset{\text{carbonic acid}}{H_2CO_3}$$

The carbonic acid dissociates:

$$\underset{\text{carbonic acid}}{H_2CO_3} \rightleftharpoons \underset{\text{hydrogen ion}}{H^+} + \underset{\text{hydrogencarbonate ion}}{HCO_3^-}$$

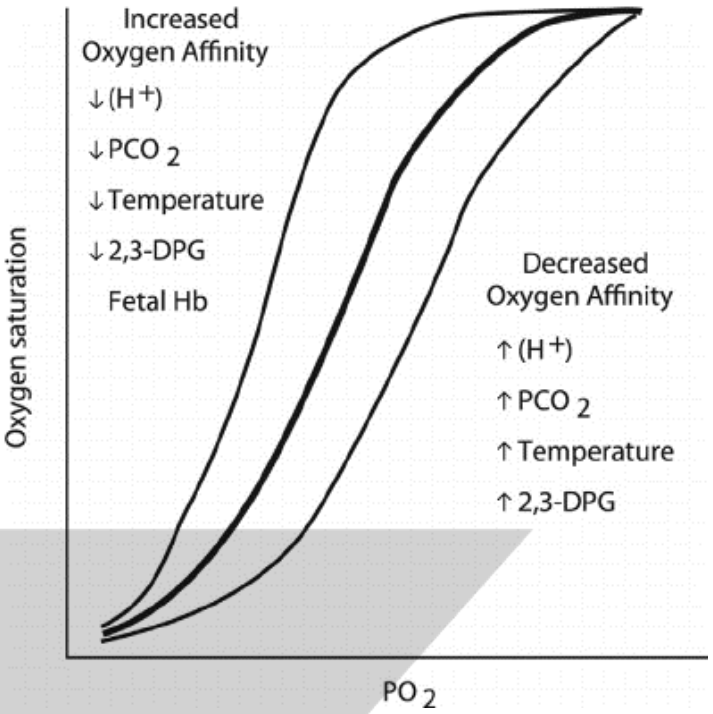
In the lungs:

- The PO2 is high, and the PCO2 is low
- CO2 in plasma diffuses from the blood into the alveoli, and oxygen diffuses into the blood from the alveoli
- Carbaminohaemoglobin dissociates to form CO2 and Hb
- Hb then picks up O2, and HHb (haemoglobinic acid) dissociates to form H+ and Hb
- The H+ ions combine with HCO3- to form carbonic acid, which dissociates to form CO2 and water (catalysed by carbonic anhydrase)

CO2 diffuses into alveoli and is breathed out

Bohr Effect:

- The presence of a high pCO_2 causes Hb to release oxygen
- High pCO_2 is found in actively respiring tissues which need oxygen
- This causes Hb to release oxygen even more readily than it would otherwise



Chloride Shift:

- When the HCO3- ions leave the red blood cells, there is a risk of developing a net positive charge inside the cell
- To prevent this from happening, Cl- ions (chloride ions) move inside the red blood cells, maintaining electrical neutrality and balancing the charges

Adaptations of Red Blood Cells:

- **Biconcave shape:** Increases surface area: volume ratio for more efficient diffusion of oxygen
- Contains **haemoglobin** to transport oxygen
- **Nucleus, mitochondrion, ER absent**: More room for Hb, thus more o2 carrying capacity.
- **Thin outer membrane** to allow oxygen to diffuse easily
- **Very small (7μm):** No haemoglobin molecule within the cell is far from the cell surface membrane. Thus, oxygen is exchanged quickly with outer fluid. Also, capillaries are 7μm allowing RBC to squeeze through.
- **Flexible:** Capillaries narrower than 7μm cause RBC to deform (as they have a specialized cytoskeleton) and return to shape in venules

1.5. External and Internal Structure of the Heart

External Structure:

- Blood vessels that leave the heart are the Aortic arch and pulmonary artery
- Blood vessels that enter the heart are the superior vena cava, the inferior vena cava and the pulmonary vein
- **The left and right sides of the heart are** separated by the septum
- The muscle from which the heart is made is called the **cardiac muscle**

Internal Structure:

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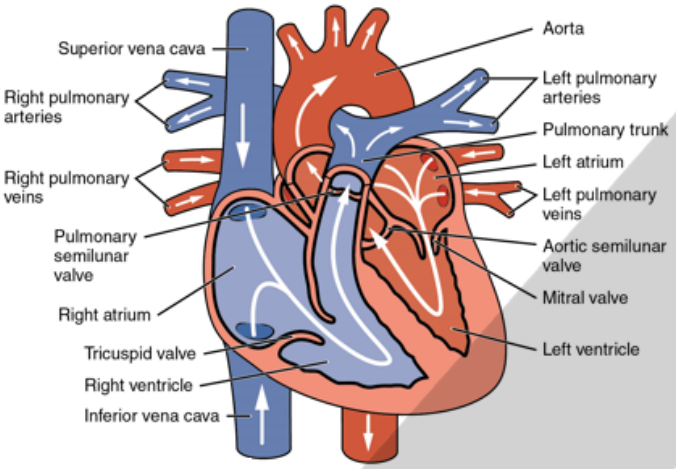
- The human heart has four chambers:
- Atria: 2 upper chambers; they are thin-walled and receive blood at low pressure
- Ventricles: 2 lower chambers are known as ventricles; they are thick-walled, receive blood from atria and pump it out through arteries
- The left ventricle has a thicker muscular wall, as it has to pump blood into the systemic circuit, which has a higher resistance to blood flow than the pulmonary circuit; the systemic circuit is also longer and requires more pressure.
- Ventricles have thicker walls than atria because they pump blood outside the heart, whereas atria pump blood to the ventricles only a short distance away.
- The atria and ventricles have valves between them (Atrioventricular valve)
- On the left, the atrioventricular valve is called 'bicuspid' or mitral
- On the right, the atrioventricular valve is called 'tricuspid'
- There are valves between the ventricles and the arteries as well, called the 'semi-lunar' valves



1.6. Cardiac Cycle

Atrial systole:

- This occurs when muscles in the atrial walls contract and blood passes on to the ventricles
- 70% of the blood flows passively down to the ventricles
- The bicuspid and tricuspid valves open while the semilunar valves are closed

Ventricular systole:

- 0.1 seconds after the atria contract, the ventricle walls contract as well, increasing the blood pressure and pushing it out of the heart
- The blood passes through the aorta and pulmonary arteries
- The semilunar valves open, and the bicuspid and tricuspid valves are closed

Ventricular diastole:

- This lasts for about 0.3 seconds; the ventricles relax, and the pressure falls below that in the arteries
- The higher pressure in the arteries pushes against the semilunar valves, shutting them

Diastole:

- All muscles of the heart relax, and the pressure inside the ventricles gets lower than in the atria
- When this happens, most of the blood starts to flow from the atria to the ventricles even though the atria are not contracting.
- However, the atria contract towards the end to push out the last bit of blood into the ventricles, and the cycle begins again.

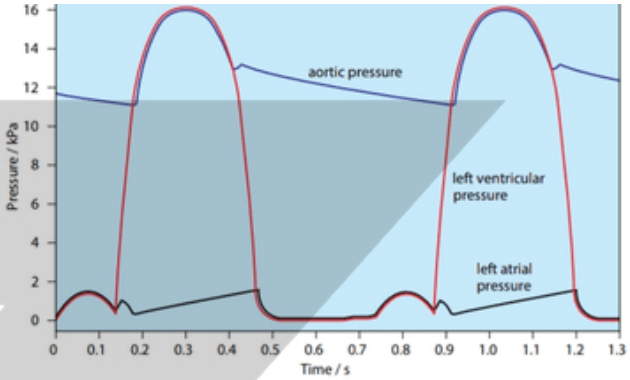


Figure 8.28 Pressure changes in the left side of the heart during the cardiac cycle.

1.7. Control of Heartbeat

- The cardiac cycle begins in the right atrium
- There is a specialised patch of muscles in the wall of the right atrium known as the sinoatrial node.
- Cells at the SAN set the rhythm for all cardiac muscle cells to beat as they send electrical impulses to the rest of the atria.
- This causes atrial contraction.
- The electrical impulses do not pass down to the ventricles
- Instead, a second node (atrioventricular node) picks up the electrical impulses from the atria
- The AV node causes the impulse to be delayed by 0.1 seconds, so the atria contract before the ventricles, giving the ventricles time to fill up with blood
- The impulse swiftly moves down to the septum of the heart, along fibres called Purkyne tissue
- Once the impulse arrives at the base of the ventricles, it moves outwards and upwards through ventricular walls
- This causes the ventricle to contract
- This is important as the ventricles must push blood upwards into the Aorta and the pulmonary artery.

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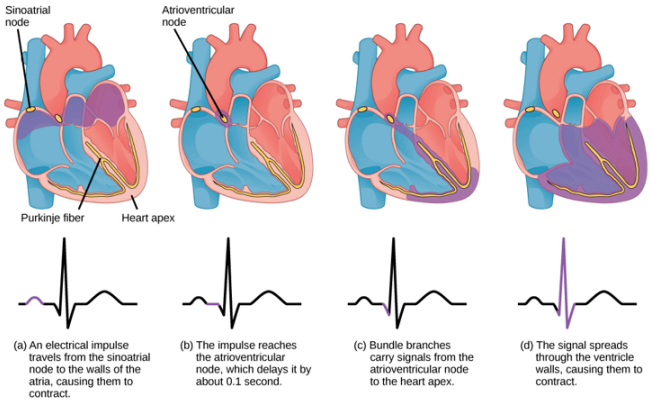

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